Efficiency and Caching

Computer Science Mentors 61A

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1 Efficiency

Efficiency: avoiding repeated work

A cache stores results so a later request can reuse them. Memoization is caching a function's return value for each input. On a *cache miss*, compute and store the result; on a *cache hit*, return the stored result.

```
def memo(f):
    cache = {}
    def remembered(x):
        if x not in cache:
            cache[x] = f(x)
        return cache[x]
    return remembered
```

Each call to `memo` creates a new dictionary. The returned function keeps that dictionary across calls. Putting `cache = {}` inside `remembered` would discard earlier results on every call. Use membership testing: a stored result of 0 or `False` is still a cached result. This version requires hashable inputs, such as integers.

Caching works when the same input should produce the same result. A cache hit skips the original function body, including its printing or other side effects. If a result depends on changing external state, reusing it can be incorrect. Caching also uses memory to store results.

Fibonacci and repeated calls

Tree-recursive Fibonacci repeats calls on smaller inputs. For example, computing `fib(5)` needs the results for inputs 4 and 3. Computing the result for input 4 needs input 3 again. With memoization, every recursive call checks a shared dictionary keyed by the argument. Each `fib(k)` is computed only once; later requests are just lookups. Saving only the final answer to the outermost call would not prevent repeated work within that computation.

List operations differ in cost

- **Fast list-building choices:** `s.append(x)` adds to the end without shifting existing elements. List comprehensions also build results efficiently, evaluating their expression once per element. Appending occasionally needs more storage, but does not copy the whole list every time.
- **Fast replacement:** `s[k] = k * k` replaces the element at an existing index without shifting other elements or copying the list. It does not grow the list: index `k` must already be valid.
- **Slow when repeated:** `s.insert(0, x)` shifts existing elements to make room at the front. `s = s + [x]` creates a new list, copying the growing old list every time.

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Explain what work is repeated or avoided, rather than giving a runtime formula.

1. Consider tree-recursive Fibonacci for non-negative integers.

```
def fib(n):
    if n < 2:
        return n
    return fib(n - 1) + fib(n - 2)
```

   (a) Draw the calls made by `fib(4)`. Which inputs occur more than once, and what does the original call return?

   (b) Suppose every call checks the same cache and saves its result. What should the keys and values be? When the second request for `fib(2)` occurs, what work is avoided?

2. Complete `make_fib` so the returned function remembers Fibonacci results across calls. Assume non-negative integer inputs.

```
def make_fib():
    cache = {0: 0, 1: 1}
    def fib(n):
        if _____________________________:
            cache[n] = _____________________________
        return _____________________________
    return fib
```

(a) Execute these calls in order. Give each return value and the new keys added to the cache during each call.

```
f = make_fib()
f(5)
f(5)
f(6)
```

(b) Why is the dictionary created outside `fib`? Would `g = make_fib()` share `f`'s saved results?

3. Compare these ways of building a list of greetings from a list of names.

```
def greetings_a(names):
    return ['Hi, ' + name for name in names]

def greetings_b(names):
    s = []
    for name in names:
        s.append('Hi, ' + name)
    return s

def greetings_c(names):
    s = []
    for name in names:
        s = s + ['Hi, ' + name]
    return s

def greetings_d(names):
    s = []
    for name in names:
        s.insert(0, 'Hi, ' + name)
    return s
```

(a) What does each function return for `names = ['Ada', 'Bo', 'Cam']`?

(b) Which two avoid repeatedly copying or shifting the growing list? Explain the extra work in the other two, without runtime notation. Focus on building the list; all four construct the same greeting strings.

(c) Rewrite `greetings_d` with a list comprehension that returns the same elements in the same order, without inserting at the front.

2 Optional Midterm Review

4. [Exam Level] Implement the `Poll` class and the `tally` function, which takes a choice `c` and returns a list describing the number of votes for `c`. This list contains pairs, each with a name and the number of times `vote` was called on that `choice` at the `Poll` with that name. Pairs can be in any order. Assume all `Poll` instances have distinct names. Hint: the dictionary `get(key, default)` method returns the value for a `key` if it appears in the dictionary and `default` otherwise. [Adapted from CS61A Fa18 Midterm 2 Q5(a)]

```
class Poll:
    s = []

    def __init__(self, n):

        self.name = ____________________________________________________

        self.votes = {}

        ____________________________________________________________________

    def vote(self, choice):

        self._____________________________ = _____________________________________


def tally(c):
    """Tally all votes for a choice c as a list of (poll name, vote count)
       pairs.
    >>> a, b, c = Poll('A'), Poll('B'), Poll('C')
    >>> c.vote('dog')
    >>> a.vote('dog')
    >>> a.vote('cat')
    >>> b.vote('cat')
    >>> a.vote('dog')
    >>> tally('dog')
    [('A', 2), ('C', 1)]
    >>> tally('cat')
    [('A', 1), ('B', 1)]
    """

    return ____________________________________________________________
```

5. Given a list of integers `lst`, return the maximum sum of a subset of size `n`. If `n` is greater than or equal to the length of `lst`, just return the sum of the elements `lst`.

```
def max_subset_sum(lst, n):
    """
    >>> max_subset_sum([1, 2, 3, 4], 2)
    7
    >>> max_subset_sum([1, 4, 2, 0, 6], 3)
    12
    """

    if _____________________________:

        _____________________________

    elif _____________________________:

        _____________________________

    with_elem = _____________________________________ + __________

    without_elem = ___________________________________________

    return ____________________________________________________
```

6. What Would Python Display? Expressions are run in the same environment.

```
class SkipMachine:
    skip = 1
    def __init__(self, n=2):
        self.skip = n + SkipMachine.skip

    def generate(self):
        current = SkipMachine.skip
        while True:
            yield current
            current += self.skip
            SkipMachine.skip += 1

p = SkipMachine()
twos = p.generate()
SkipMachine.skip += 1
twos2 = p.generate()
threes = SkipMachine(3).generate()
```

(a) `next(twos)`

(b) `next(threes)`

(c) `next(twos)`

(d) `next(twos)`

(e) `next(threes)`

(f) `next(twos2)`